Neelima Gurram

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Summary

M.S. Computer Science graduate (May 2026) with experience building recommender systems, LLM-powered applications, and distributed ML pipelines. Proficient in Python, PyTorch, LangChain, and AWS. Skilled at working with large-scale datasets and applying machine learning to drive user-facing product impact.

Education

Wright State University *May 2026*
Master of Science in Computer Science — GPA: 4.0 / 4.0 Dayton, Ohio

Experience

Data Science Intern *Jan 2024 – Apr 2024*
Brain O Vision (AICTE), Hyderabad, India

- Built scalable NLP pipelines (Python, scikit-learn, NLTK, spaCy) to classify 50,000+ social media posts; improved accuracy to 87% and reduced false positives by 30% via feature engineering and model optimization.
- Developed Flask API workflows for model inference and automated alert generation, cutting manual review workload by 40%.

Cloud Computing Intern *Jun 2023 – Aug 2023*
Corizo (MSME), Bengaluru, India

- Deployed AWS infrastructure (EC2, S3, IAM, Lambda, VPC), automated provisioning via shell scripting, and built an Amazon Lex AI chatbot for customer support workflows.

Projects

NextRead — Book Recommender System | *Python, SVD, Collaborative Filtering, scikit-learn, Surprise*

- Developed a personalized book recommendation engine on the Book-Crossing dataset comprising 278K users, 271K items, and 1.1M ratings — comparable in scale to real-world content ranking systems.
- Implemented and benchmarked memory-based collaborative filtering (user-based and item-based cosine similarity) against model-based matrix factorization using Singular Value Decomposition (SVD).
- SVD reduced RMSE from 7.94 to 1.64 by learning latent user preference and item attribute factors, demonstrating superior handling of sparse user-item interaction matrices.
- Analyzed scalability trade-offs between $O(n^2)$ pairwise similarity computation and SVD dimensionality reduction; evaluated cold-start behavior and generalization to unseen users and items.

Adversarial Robustness of GNNExplainer (M.S. Thesis) | *PyTorch Geometric, Graph Neural Networks*

- Investigated the vulnerability of GNN explanation methods to adversarial perturbations on graph-structured data; designed targeted attack strategies to quantify explanation instability.
- Built end-to-end experimentation pipelines to measure robustness and explanation drift across multiple benchmark graph datasets.
- Results highlighted critical failure modes in post-hoc GNN explainability, contributing to safer and more reliable graph ML deployment.

LLM Document Q&A System | *LangChain, Flask, GPT, FAISS, RAG*

- Built an end-to-end LLM-powered document retrieval and question-answering platform using LangChain, OpenAI GPT models, and FAISS vector databases.
- Implemented retrieval-augmented generation (RAG) pipelines for semantic search, enabling accurate querying across large multi-document PDF collections with structured analytical outputs.

Distributed SGD Optimization Study | *PyTorch, TensorFlow, Dask, Ray*

- Benchmarked distributed ML training strategies across PyTorch DDP, TensorFlow MirroredStrategy, Dask, and Ray Train on large-scale datasets.
- Designed adaptive gradient synchronization techniques that improved training scalability by 25% and reduced straggler-induced delays by 30% compared to synchronous baselines.

Technical Skills

Programming: Python, Java, JavaScript, SQL

AI / ML: Recommender Systems, Collaborative Filtering, Matrix Factorization (SVD), LangChain, LLMs, NLP, RAG, PyTorch, TensorFlow, scikit-learn, GNNs, PyTorch Geometric, Neural Networks

Backend / Cloud: Flask, FastAPI, React, AWS (EC2, S3, Lambda, IAM, VPC), Git, Linux, CI/CD

Data / Distributed: MySQL,Pandas, NumPy, Matplotlib, Dask, Ray, Distributed Training